CEN-JHIH LI

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Machine Learning PhD focused on efficient LLM training and inference, with research in sparse fine-tuning, memory-efficient inference, and in-context learning. Experienced in PyTorch/JAX transformer implementations from scratch, quantized LLM fine-tuning, and multi-GPU training.

EXPERIENCE AND EDUCATION

PhD in Computer Science, University of Utah *Aug 2022 – Expected 2027-2028 (flexible)*

- Focus: Efficient foundation models, including memory-efficient adaptation, model compression, LLM inference, in-context learning, and understanding model behavior and representations.
  Advisor: Prof. Aditya Bhaskara
- **Selected Projects:**
  - **Sparse Fine-Tuning & Quantized LLM Adaptation (TMLR 2025)**: Developed structured sparse fine-tuning for quantized LLMs, reducing additional fine-tuning memory by 20–50% versus baselines and eliminating quantization error on selected rows at initialization. Implemented LoRA/QLoRA fine-tuning and LoftQ-style initialization using quantization residuals.
  - **LLM Inference & Hardware-Aware KV Token Selection (Ongoing)**: Investigating efficient token selection for sparse attention and KV-cache eviction using a small subset of key frequency bands as an indexer without storing a separate per-token index. Implementing and profiling kernels in PyTorch and Triton.
  - **Transformer Training & GPU Efficiency**: Implemented custom Transformer models and multi-GPU training/inference with data parallelism, tensor sharding, gradient accumulation, layer-selective activation checkpointing, and all-gather, all-reduce, and reduce-scatter collectives.

Research Assistant, Institute of Statistical Science, Academia Sinica *Jul 2020 – Jul 2022*

- Developed a robust aggregation method for federated learning using statistical approaches.
  Supervisor: Prof. Su-Yun Huang
- Research on robust federated learning won the ISSAS Poster Competition and was presented at the CSA-KSS-JSS International Statistical Conference.

Data Scientist at Data-on-Demand *Dec 2018 – Jan 2020*

- Applied machine learning and data analysis techniques to improve commercial decision-making for a fitness chain and an insurance company (Allianz, Shanghai).

Mandatory Military Service, Taiwan *Oct 2017 – Oct 2018*

MS in Statistics, National Chengchi University *Sep 2015 – Jul 2017*

- Thesis: Handling missing-not-at-random (MNAR) data in online product reviews using multiple imputation.
  Advisors: Prof. Kwei Tang and Prof. Tsung-Chi Cheng

BS in Applied Mathematics, National Chiao Tung University *Sep 2011 – Jun 2015*

Additional Honors & Credentials: Graduate Fellowship (University of Utah); third place, FDT Foreign Exchange Futures Investment Competition; Society of Actuaries Exams P and FM; Taipower Scholarship ($< 10\%$ acceptance rate) and Cathay Outstanding Student Scholarship.

SKILLS

- **Languages & Frameworks**: C++, Python, PyTorch, Triton, JAX, Hugging Face Transformers
- **ML Systems**: LLM training/inference profiling, roofline and bottleneck analysis, distributed training and inference, model compression, KV caching, and memory optimization
- **Computing**: HPC and GPU cluster environments, distributed-model scaling and compute, memory-bandwidth, and communication tradeoffs

Publications, ongoing research, and open-source contributions on the following page

PUBLICATIONS AND ONGOING RESEARCH

- **Li, C.-J.**, Awal, T., Sundar, H., & Bhaskara, A. *Pretraining Design Drives Gains in Zero-Shot PDE Rollouts*. Under review at ICLR 2027.
- Awal, T., **Li, C.-J.**, Bhaskara, A., & Sundar, H. *Tunable Non-linear Manifold ROMs for PDEs via Matrix-Free Petrov–Galerkin Projection*. Under review at ICLR 2027.
- **Li, C.-J.**, Chaudhary, S., & Bhaskara, A. *Revisiting What Makes a Good Context Subset: A Sparse Latent Basis Perspective*. Under review at ACL Rolling Review (ARR), 2026.
- **Li, C.-J.** & Bhaskara, A. (2025). *An Efficient Sparse Fine-Tuning with Low Quantization Error via Neural Network Pruning*. Transactions on Machine Learning Research (TMLR), 2025. OpenReview Link.
- **Li, C.-J.** & Bhaskara, A. (2025). *An Efficient Row-Based Sparse Fine-Tuning with Low Quantization Error*. Proceedings of the 3rd Efficient Systems for Foundation Models (ES-FoMo) Workshop at ICML 2025, PMLR 267. OpenReview Link
- **Li, C.-J.**, Huang, P. H., Ma, Y. T., Hung, H., & Huang, S. Y. (2022). *Robust Aggregation for Federated Learning by Minimum γ-Divergence Estimation*. Entropy, 24(5), 686. GitHub Repo.
- **Ongoing Research**:
  - Studying how context position and ordering affect downstream query performance in in-context learning.
  - Investigating context utility in long-context language models to improve KV-cache selection during autoregressive inference.

OPEN-SOURCE CONTRIBUTIONS

- **FastChat** - Integrated the google.genai API into a widely used LLM-as-a-judge evaluation toolkit, adding support for Gemini 2.5 as a judge model. PR #3751
- **OpenICL** - Updated compatibility with recent versions of Hugging Face Transformers. PR #31